Supplemental Table S1. Evidence table with included studies and corresponding study characteristics and outcomes.

Author (Year)	Study Groups (n)	Mean Age (y)	Additional surgeries	BMI	Defect Characteristics	Measures	Follow-up	Results
Bone marrow stimulation								
Kreuz (2006)[44]	**Therapy:** All patients underwent MF (70) **Subgroup analysis on age and defect location:** Group 1a: femoral condyles ≤40 y (16) Group 1b: femoral condyles >40 y (16) Group 2a: tibia ≤40 y (4) Group 2b: tibia >40 y (7) Group 3a: patellofemoral ≤40 y (16) Group 3b: patellofemoral > 40 y (11)	**Overall:** 39.5 **Group 1a:** 29.9 **Group 1b:** 47.5 **Group 2a:** 26.2 **Group 2b:** 47.4 **Group 3a:** 34.1 **Group 3b:** 49.4	No previous surgeries reported No concomitant surgeries reported	**Group 1a:** 23.3 **Group 1b:** 27.0 **Group 2a:** 24.3 **Group 2b:** 26.0 **Group 3a:** 24.2 **Group 3b:** 28.2	**Size and Outerbridgde classification:** Group 1a: 2.13 cm^2, 3.75 Group 1b: 1.97 cm^2, 3.75 Group 2a: 2.19 cm^2, 3.75 Group 2b: 2.18 cm^2, 3.57 Group 3a: 2.38 cm^2, 3.88 Group 3b: 2.39 cm^2, 3.82	Cincinnati knee score, ICRS scoring system, MRI 18 and 36 mo, Cincinnati score	36 mo (NOS)	**Cincinnati score:** Group 1a and 3a > group 1b and 3b (P<0.01) Between 18 and 36 mo, ICRS scores deteriorated in patients >40 y *(P<0.05)* **MRI defect filling:** Patients ≤40 y > patients >40 y *(P<0.05)*
Sansone (2014)[74]	**Therapy:** All patients underwent bone marrow stimulating abrasion arthroplasty 53) **Subgroup analysis on age:** Patients aged <50 y (25) vs ≥50 y (28)	**Overall:** 45.9	None (part of the exclusion criteria)	Not provided	**Defect size:** **<50 y:** 7.5% > $4cm^2$ **≥50 y:** 28.3% > $4cm^2$ (P=0.022) **Defect grade:** All defects grade IV Outerbridge (NOS)	KSS, WOMAC, Survival defined as requiring second surgery	20.1 y	**KSS score:** 84% in patients <50 y vs 53.6% ≥50 y (*P=0.046*) **Survival:** 89.5% in patients aged <50 y vs 55.7% in ≥50 y **Arthroplasties:** 1 in patients aged <50 y vs.12 in ≥50 y
Gille (2013)[29]	**Therapy:** All patients underwent AMIC (57) **Subgroup analysis on age:** Group A: 17 to 32 y (17) Group B: 33 to 46 y (27) Group C: 47 to 65 y (13)	**Overall:** 37.3	**Previous surgeries:** Diagnostic arthroscopies (10), partial meniscectomies (5), shavings (16), drilling or microfracture (4) **Concomitant surgeries:** Patella realignments (2), corrective osteotomies (3), partial meniscectomies (6) and ACL reconstruction (1)	Not provided	**Mean defect size:** 3.4 cm^2 **Defect grade:** 35% Outerbridge III and 65% Outerbridge IV	Lysholm, VAS	2 y (NOS)	**Lysholm score:** No statistical differences between the three age groups (*P=0.085)*
Cell-based techniques								
Nehrer (2006)[65]	**Therapy:** All patients underwent ACI-m (36) **Subgroup analysis on age:** Patients aged <30 y (17) vs >30 y (19)	**Overall:** 33.0	No previous surgeries reported No concomitant surgeries reported	Not provided	**Mean defect size:** Range 1.5-8.0 cm^2 (NOS)	Lysholm Cincinnati score, IKDC, form, ICRS grading	36 mo (NOS)	**Lysholm and IKDC:** Significant improvement compared to preoperative in patients <30 y vs no significant improvement in patients >30 y *(P<0.01)*
Krishnan (2006)[45]	**Therapy:** All patients underwent ACI-c (199) **Subgroup analysis on age:** -Group 1: <20 y (28) -Group 2: 21-40 y (137) -Group 3: >41 y (34)	Not provided	Up to three previous surgeries (NOS) No concomitant surgeries reported	Not provided	**Mean defect size:** 4.9 $cm^{2,}$	Modified Cincinnati, ICRS grading, Histology	4 y (NOS)	**Cincinnati score:** group 1: 76.7 vs group 2: 60.8 vs group 3: 55.0 *(P < 0.001)* **Good/Excellent results:** group 1: 85.7% vs group 2: 64.2% vs group 3: 55.9% *(P = 0.04)*
Niemeyer (2010)[67]	**Therapy:** All patients underwent ACI-c (74) **Matched-pair analysis:** Patients aged <40 y (37) *vs* ≥40 y (37)	**Patients <40:** 31.1 **Patients ≥40:** 44.8	No previous surgeries reported No concomitant surgeries reported	**<40 y:** 24.5 **≥40 y:** 23.2	**Defect size:** Not significantly different between the groups (NOS). **Defect grade:** ICRS grade III or IV	IKDC, Lysholm, Cincinnati sports activity scale, TAS	24 mo (NOS)	**All outcomes:** <40 y = patients ≥40 y *(P<0.05)*

Author (Year)	Study Groups (n)	Mean Age (y)	Additional surgeries	BMI	Defect Characteristics	Measures	Follow-up	Results
Brix (2014)[9]	**Therapy:** All patients underwent ACI-m **Subgroup analysis on age:** Patients aged <40 y (22) *vs* >40 y (5)	**Overall:** 33.4	**Previous surgeries: I**n 75.5% of total group **Concomitant surgeries:** HTO, ACL reconstruction (NOS)	**Overall:** 24.5	**Mean defect size:** 4.4 cm^2 **Defect grade:** ICRS grade III or IV	Failure, Survival, Lysholm, IKDC	5.0 y	**Lysholm:** Δ 25.8 in patients <40 y *vs* Δ 5.6 in patients >40 y *(P=0.045)*
Filardo (2014)[25]	**Therapy:** All patients underwent ACI-m (142)	**Overall:** 29.2	**Previous surgeries:** In 71/142 patients including: meniscectomy, ACL reconstruction, etc. **Concomitant procedures:** In 54/142 patients including: meniscectomy, ACL reconstruction, etc.	**Overall:** 24.0	**Mean defect size:** 2.3 cm^2 **Defect grade:** ICRS grade III or IV	IKDC, EQ, VAS, TAS	7 y (NOS)	**IKDC score:** Improvements of 80% were only observed in patients <40 y and not in patients >40 y (no statistical test)
Filardo (2017)[24]	**Therapy:** All patients underwent ACI-m (157) **Subgroup analysis on age:** Patients aged <40 y (122) *vs* >40 y (35)	**Patients <40:** 25.6 **Patients >40:** 46.0	**Previous surgeries:** Patients aged <40 y 53.3% *vs* >40 y 71.4% *(NS)* **Concomitant surgeries:** Realignment or ligament reconstruction 50.8% in patients aged <40 y *vs 42.9% in patients* >40 y 42.9% *(NS)*	**<40 y:** 24.0 **>40 y:** 26.4 *(P<0.0005)*	**Mean defect size:** **<40 y:** 2.7 cm^2 **>40 y:** 3.1 cm^2 **Defect grade:** ICRS grade III or IV	Failure, IKDC, TAS	10 y (NOS)	**IKDC scores:** Patients <40 y < patients >40 y *(P=0.007)*
Gobbi (2014)[30]	**Therapy:** All patients underwent BMAC (25) **Subgroup analysis on age:** Patients aged <45 y vs >45 y (multivariate analysis)	**Overall:** 46.5	**Previous surgeries:** Such as HTO are mentioned but no details are provided **Concomitant surgeries:** Knee stabilizing surgeries are mentioned but no details are provided	**Overall:** 24.4	**Mean defect size:** 8.3 cm^2 **Defect grade:** ICRS grade IV (inclusion criterium)	ROM, VAS, IKDC, KOOS, Lysholm, Tegner, Marx	41.3 mo	**Improvements at final follow-up (NOS):** Patients <45 y = patients >45 y (NS)
Gobbi (2017)[31]	**Therapy:** All patients underwent HA-BMAC (40) **Study groups:** Patients aged >45 y (20) vs <45 y (20)	**Patients <45:** 36.6 **Patients >45:** 50.0	No previous surgeries reported **Concomitant procedures:** HTO: 9 in >45 vs 2 in <45; ACL reconstruction: 3 in >45 vs 1 in the <45; Lateral release (patella): 1 in >45 vs 3 in <45; Fulkerson: 1 in >45 vs 5 in the <45; Meniscectomy: 0 in >45 vs 1 in <45 (no statistics)	Not provided	**Mean defect size:** **>45 y:** 8.5 cm^2 **<45 y:** 9.8 cm^2 **Defect grade:** ICRS grade IV	VAS, IKDC, KOOS, Tegner, MRI	**>45 y:** 48.7 mo **<45 y:** 52.3 mo	**Tegner, KDC, KOOS: N**o significant differences in improvement between the age groups **VAS: I**mproved more in patients >45 y vs <45 y (*P=0.046)* **Complete/near complete defect filling:** 81% in patients >45 y vs 71% in patients <45 y (no statistics provided) **Integration of graft with adjacent cartilage:** 93.7% in patients >45 y vs 93% in patients <45 y (no statistics provided)
Gobbi (2019)[32]	**Therapy:** All patients underwent BMAC (23) **Subgroup analysis on age:** Patients aged ≤45 y vs >45 y	**Overall:** 48.5	No previous surgeries reported **Concomitant procedures:** HTO (8); ACL reconstruction (3); Lateral release (patella) (2); TTO (3)	**Overall:** 24.4	**Mean defect size:** 6.5 cm^2 **Defect grade:** ICRS grade IV (inclusion criterium)	IKDC, KOOS, Tegner, VAS	8 y (Mean)	**IKDC, KOOS, Tegner and VAS pain score:** Patients ≤45 y = patients >45 y (*P >0.05)* Significant negative correlation between age and IKDC, Tegner and KOOS scores.

Author (Year)	Study Groups (n)	Mean Age (y)	Additional surgeries	BMI	Defect Characteristics	Measures	Follow-up	Results
Bone-based techniques								
Levy (2013)[49]	**Therapy:** All patients underwent Shell OCA **Subgroup analysis on age:** Patients aged <30 y vs ≥30 y (multivariate analysis)	**Overall:** 32.8 (85% younger than 45)	**Previous surgeries:** Received by the majority of patients, some of them more than one. Chondroplasty was performed in 73.5% and bone marrow stimulation in 40% No concomitant surgeries reported	**Overall:** 25.7	No defect characteristics provided	Modified Merle d'Aubigné-Postel scale, IKDC, KSS, Failure	13.5 y (Median)	Graft failure: A 3.5 odds ratio in patients ≥30 y when compared to the reference group (logistic regression analysis)
Frank (2017)[26]	**Therapy:** All patients underwent OCA (170) **Subgroup analysis on age:** Patient aged <40 y (115) *vs* ≥ 40 y (55)	**Patients <40:** 27.6 **Patients >40:** 44.9	No previous surgeries reported **Concomitant treatments:** Distal femoral osteotomy <40 y: 8% vs >40 y: 0% *(P=0.028)*. Lateral MAT was significantly more performed in in patients <40 *(P<0.0001)*	**<40 y:** 25.9 **>40 y:** 28.1 *(P=0.18)*	**Mean defect size:** 4.5 cm^2	Reoperations, Complications, IKDC, KOOS, WOMAC, SF-12.	5.0 y	**Re-surgery rate, time to re-surgery, failure rate:** patients ≥40 y = patients <40 y *(NS)* **KOOS symptoms subscore:** Patients ≥40 y > patients <40 y *(P=0.015)*
Anderson (2018)[4]	**Therapy:** All patients underwent OCA (80) **Subgroup analysis on age:** Patients aged ≤39 y (42) vs ≥40 y (38)	**Patients ≤39:** 27.2 **Patients ≥40 y:** 52.3	No previous surgeries reported **Concomitant treatment:** Patients aged ≤39 y 33% vs patients ≥40 y 24% *(P=0.459)* (HTO, Tibial Tubercle Osteotomy, Loose body removal, ACL reconstruction, Meniscectomy, Lateral release or MAT)	**≤39 y:** 26.55 **≥40 y:** 28.29 *(P=0.085)*	**Mean defect size:** **≤39 y:** 5.22 cm^2 **≥40 y:** 5.91 cm^2 *(P=0.476)* **Defect grade:** ICRS grade IV	KOOS, IKDC, VR-12	**≤39 y:** 33.8 mo **≥40 y:** 44.5 mo	**All outcomes:** Patients ≤39 y = patients ≥40 y *(NS)*
Martinez-Carranza (2020)[56]	**Therapy:** All patients received an Episurf customized metal implant (30); 26 on the medial femoral condyle and 4 in the trochlea	47 (SD 8.4)	Concomitant injuries were part of the exclusion criteria	27	**Mean defect size:** 256 mm^2 **Defect grade:** 18/30 were ICRS grade 4	KOOS, EQ5D, VAS	**Mean** 55 mo KOOS, EQ5D and VAS 24 mo	**Reoperation rate:** 17%, 1/30 patients had conversion to an hemiarthroplasty **KOOS, EQ5D and VAS**: significant improvements at 24 mo > preoperative
Holz (2020)[39]	**Therapy:** All patients received an Episurf customized metal implant (75); 60 on the medial femoral condyle, 10 in the trochlea and 5 on the lateral condyle	48 (SD 8.3)	**Previous surgeries**: 48/75; 31 MF, 5 debridement, 5 OAT, 4 ACI, 3 Trufit. **Concomitant surgeries:** no details provided	28	**Mean defect size:** Medial femoral condyle: <3cm^2 Lateral femoral condyle: 3-4 cm^2 Trochlea: >4 cm^2 (*NOS*) **Defect grade:** ICRS III–IVb (by inclusion, *NOS*)	KOOS, VAS, failure	24 mo	**KOOS and VAS:** all scores improved significantly at 24 mo > preoperative Minimal clinically important difference in KOOS was maintained during follow-up **Failure:** 2 revisions (2.5%) at final follow-up

Author (Year)	Study Groups (n)	Mean Age (y)	Additional surgeries	BMI	Defect Characteristics	Measures	Follow-up	Results
Studies including different treatments								
de Windt (2009)[16]	**Therapy:** ACI (25) and MF (30) **Subgroup analysis on age:** ACI: <30y (11) and >30y (14) MF: <30y (13) and >30y (17)	**Overall:** 35	No previous surgeries reported No concomitant surgeries reported	Not provided	**Mean defect size:** **ACI:** 3.25 cm^2 **MF:** 2.60 cm^2	KOOS	36 mo	**KOOS:** patients <30 y > patients >30 y (*P<0.05)*
Nejadnik (2010)[66]	**Therapy:** BMAC (36) vs ACI (36) **Subgroup analysis on age:** Patients aged <45 y (35) vs ≥45 y (37)	**BMAC:** 44.0 **ACI:** 42.5	No previous surgeries reported. **Concomitant surgeries:** Patella realignment: 6 in ACI, 5 in BMAC; HTO: 5 in BMAC; Partial meniscectomy: 1 per group; ACL reconstruction: 1 per group	Not provided	**Mean defect size:** **BMAC:** 4.6 cm^2 **ACI:** 3.6 cm^2 (*P=0.270)*	IKDC, SF 36, Lysholm, TAS	24 mo (NOS)	**IKDC: P**atients <45 y = ≥45 y in ACI and BMAC **TAS:** in ACI patients <45 y > ≥45 y; in BMAC patients <45 y = ≥45 y
Pascual-Garrido (2016)[69]	**Therapy:** HemiCAP focal femoral condyle resurfacing prosthesis (32) vs MF (15), OCA (10), OAT (2), debridement (2) or ACI (1)	**Intervention:** 44.5 **Control:** 47.9	**Previous surgeries:** 84% of patients in intervention group had undergone previous surgeries vs 63% of patients in the control group No concomitant surgeries reported.	**Intervention:** 26.7 **Control:** 30.4 (P=0.03)	**Defect grade:** ICRS grade IV	Need for intervention, WOMAC, SF-12, Satisfaction, Adverse events	**Intervention:** 2.0 y **Control:** 2.6 y (P=0.0002)	**Clinical success:** Intervention: 75% vs control 53% (*P* <0.001) **WOMAC pain domain:** **I**ntervention > control (*P*=0.03) **Good/Excellent satisfaction:** Intervention: 91% vs control: 66% (*P*<0.001)
Nathwani (2017)[64]	**Therapy:** BioPoly RS Partial Resurfacing Knee Implant vs historical MF cohorts (Saris 2014: mean patient age 32.9 y [72]; Saris 2008: mean patient age 33.9 [61]; Rotterud 2016: mean patient age 35 y [88]; Cole 2011: mean patient age 33 y [9]) **Subgroup analysis on age:** Patients aged ≤40 y *vs* >40 y	**Overall:** 42.7 (60.6% >40)	**Previous surgeries: P**erformed in 75.8% of patients.	**Overall:** 26.7	**Mean defect size:** 2.7 cm^2 **Defect grade:** ICRS grade IV grade II-IV	KOOS, VAS, SF-36, TAS	2 y (NOS)	**KOOS, VAS, SF-36, TAS:** significant increased compared to preoperatively *(P < 0.025)* **KOOS, VAS, SF-36:** Patients ≤40 y = patients >40 y *(P>0.05)* **Quality of life KOOS subscore:** FKRI > all 4 used MFX studies *(P<0.025)* **Sports KOOS subscore:** FKRI > to MFX in Saris (2014) and Rotterud (2016) *(P<0.025)* **ADL KOOS subscore:** FKRI > MFX in Saris (2014) *(P<0.025)*

Abbreviations: ***ACI****: Autologous Chondrocyte Implantation;* ***ACI-c****: Autologous Chondrocyte Implantation with collagen flap;* ***ACI-m****: Autologous Chondrocyte Implantation with matrix assistance;* ***ACL****: Anterior Cruciate Ligament;* ***AMIC****: Autologous Matrix Induced Chondrogenesis;* ***BMAC****: Bone Marrow Aspirate Concentrate;* ***BMI****: Body Mass Index;* ***EQ****: EuroQol;* ***EQ5D****: EuroQol Five Dimensions Health Questionnaire;* ***FKRI****: focal knee resurfacing implant;* ***HA-BMAC****: BMAC with hyaluronic scaffold;* ***HTO****: High Tibial Osteotomy;* ***ICRS****: International Cartilage Repair Society;* ***IKDC****: International Knee Documentation Committee Subjective Knee Form;* ***KOOS****: Knee Injury and Osteoarthritis Outcome Score;* ***KSS****: Knee Society Score;* ***MAT****: Meniscus Allograft Transplant;* ***MF:*** *Microfracture* ***MRI****: Magnetic Resonance Imaging;* ***NOS****: Not Otherwise Specified;* ***OA****: Osteoarthritis;* ***OAT****: Osteochondral Autografting;* ***OCD****: Osteochondritis Dissecans;* ***ROM****: Range of Motion;* ***SF-12****: Short Form 12 questionnaire;* ***SF-36****: Short Form 36 Questionnaire;* ***TAS****: Tegner Activity Score;* ***TTO****: Tibial Tubercle Osteotomy;* ***VAS****: Visual Analogue Score;* ***VR-12****: Veterans RAND 12 Item Health Survey;* ***WOMAC****: Western Ontario and McMaster Universities Osteoarthritis Index.*